

Example: Using Karush-Kuhn-Tucker Conditions for Nonlinear Optimization Problems with Linear Constraints

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I The Problem

Consider the following optimization problem:

$$\begin{aligned} \min \quad & x_1^2 + 2x_2^2 + 4x_1x_2 \\ \text{subject to} \quad & \\ & x_1 + x_2 = 1, \\ & x_1, x_2 \geq 0 \end{aligned}$$

II Solution

The Karush-Kuhn-Tucker (KKT) conditions are both necessary and sufficient global optimality conditions when the objective function of a minimization problem is convex. When the objective function is nonconvex, the KKT conditions are necessary optimality conditions. To see if the KKT conditions are sufficient, we determine if the problem is convex. To do so, we calculate the Hessian matrix of the objective function. The Hessian matrix is given by:

$$H_f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Taking the Hessian of our objective function, A :

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$$

[†]Example problem from *Nonlinear Programming* by Amir Beck.

If A is positive semi definite, the objective function is convex. A is positive semi definite if and only if all eigenvalues are non-negative. However, a shortcut can be used since our matrix is 2x2. A will be positive semi definite if and only if:

$$\text{Tr}(A), \det(A) \geq 0$$

Calculating the trace and determinant of A gives:

$$\text{Tr}(A) = 3, \det(A) = -2$$

Therefore, A is not positive semi definite, and the function is nonconvex. However, the KKT conditions are still necessary. Next, we must find the KKT conditions of the problem. To do that, we start by taking the Lagrangian. The Lagrangian is defined as:

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = f(x) + \lambda^T(\mathbf{Ax}-\mathbf{b}) + \mu^T(\mathbf{Cx}-\mathbf{d})$$

Taking the Lagrangian of our problem:

$$\mathcal{L}(x_1, x_2, \mu, \lambda_1, \lambda_2) = x_1^2 + 2x_2^2 + 4x_1x_2 + \mu(x_1 + x_2 - 1) - \lambda_1x_1 - \lambda_2x_2$$

$$\lambda_1, \lambda_2 \in \mathbb{R}_+, \mu \in \mathbb{R}$$

Our KKT conditions are:

$$\frac{\partial \mathcal{L}}{\partial x_1} = 2x_1 + 4x_2 + \mu - \lambda_1 = 0$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = 4x_2 + 4x_1 + \mu + \lambda_2 = 0$$

$$\lambda_1x_1 = 0$$

$$\lambda_2x_2 = 0$$

$$x_1 + x_2 = 1$$

$$x_1, x_2, \lambda_1, \lambda_2 \geq 0$$

We must consider 4 possible cases of the values of the Lagrange Multiplier (λ) to find out KKT points and optimal solution.

Case 1

$$\lambda_1 = \lambda_2 = 0$$

This gives the equations:

$$2x_1 + 4x_2 + \mu = 0$$

$$4x_2 + 4x_1 + \mu = 0$$

$$x_1 + x_2 = 1$$

The solution to this system of equations is:

$$x_1 = 0, x_2 = 1, \mu = -4$$

Therefore, our KKT point is:

$$(x_1, x_2) = (0, 1)$$

Case 2

$$\lambda_1, \lambda_2 > 0$$

By our complementary slackness conditions:

$$\lambda_1 x_1 = 0$$

$$\lambda_2 x_2 = 0$$

x_1, x_2 must be zero since λ_1, λ_2 are greater than zero. This, however contradicts our constraint

$$\begin{aligned} x_1 + x_2 &= 1 \\ 0 + 0 &= 0 \neq 1 \end{aligned}$$

So, we do not obtain any KKT points from Case 2.

Case 3

$$\lambda_1 > 0, \lambda_2 = 0$$

By the complementary slackness conditions, x_1 must be zero. Turning to our constraint

$$\begin{aligned} x_1 + x_2 &= 1 \\ 0 + x_2 &= 1 \\ x_2 &= 1 \end{aligned}$$

Checking the first 2 equations in our KKT conditions

$$4 + \mu - \lambda_1 = 0$$

So we identify (0,1) which is already the KKT point we found in Case 1.

Case 4

$$\lambda_2 > 0, \lambda_1 = 0$$

Again, by our complementary slackness conditions, $x_1 = 1, x_2 = 0$. Plugging into our first two equations in our KKT conditions:

$$\begin{aligned} 2 + \mu &= 0 \\ 4 + \mu - \lambda_2 &= 0 \end{aligned}$$

The solution to this system of equations is:

$$\mu = -2, \lambda_2 = 0$$

Which satisfies our condition:

$$x_1, x_2, \lambda_1, \lambda_2 \geq 0$$

Therefore, our second KKT point is

$$(x_1, x_2) = (1, 0)$$

How do we know if these points are optimal solutions? We must consider the Weierstrass Theorem.

Let f be a continuous function defined over a nonempty and compact set $C \subseteq \mathbb{R}^n$. Then, there exists a global minimum point of f over C and a global maximum point of f over C .

Our problem minimizes a continuous function over a compact set, so it has a global optimal solution. The KKT conditions are necessary optimality conditions. Therefore, one of our KKT points is optimal. Plugging our points into the objective function:

$$f(0, 1) = 2$$

$$f(1, 0) = 1$$

Clearly, $(1, 0)$ is the optimal solution since it is the smallest value.